



RS003 Week 14 — English Worksheet

Topic: A HUD and Enemy Types · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week is about **thinking about and explaining** the code for a game that shows a **score and timer** on screen (a HUD) and has **several enemies** at once, each a different type. Each enemy is a dictionary with its own position, type, and points — aliens are worth 1 point and meteors are worth 3.

Words to know: **HUD, score, timer, list of dictionaries, enemy type**

1 · Predict 🧠

Read each piece of code. Write what you think it does. You do not run anything — just think and explain.

```
font = pygame.font.SysFont(None, 32)
score_text = font.render(f"Score: {score}", True, (255, 255, 255))
screen.blit(score_text, (10, 10))
```

These three lines put text on screen. What does the player see in the top-left corner?


```
enemies = []
for i in range(3):
    kind = random.choice(["alien", "meteor"])
    points = 1 if kind == "alien" else 3
    enemies.append({"x": random.randint(150, WIDTH - 150), "y": 120, "type": kind,
"points": points})
```

How many enemies are made? What is different about each one?


```
enemies = [new_enemy() for i in range(3)]
```

This is a shorter way to fill a list. How many enemies go into the list, and what does `new_enemy()` give back each time?


```
score = score + hit["points"]
```

Score goes up by the enemy's `points`. An alien is worth 1 and a meteor is worth 3. Which gives more points?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Write the fixed code, then explain why the original was wrong.

Bug A — The score is supposed to appear on screen, but the number never shows up.

```
score_text = font.render(f"Score: {score}", True, (255, 255, 255))
```

Hint: rendering the text is not enough — it must be put on the screen.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Hitting a meteor should give more points than an alien, but every enemy gives the same score.

```
score = score + 1
```

Hint: the score should depend on the enemy's type.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Pressing SPACE on empty space should do nothing, but the program crashes when you miss.

```
hit = get_aimed_enemy(ship_x)
score = score + hit["points"]
```

Hint: when nothing is lined up, `hit` is `None`, and `None["points"]` is an error.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working program. Then answer the questions below. You do not run it — just read it carefully and explain.

```

import pygame
import random
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()
font = pygame.font.SysFont(None, 32)

RED = (230, 60, 60)
GREY = (120, 120, 140)

def new_enemy():
    kind = random.choice(["alien", "meteor"])
    points = 1 if kind == "alien" else 3
    return {"x": random.randint(150, WIDTH - 150), "y": 120, "type": kind,
            "points": points}

enemies = [new_enemy() for i in range(3)]

def draw_enemy(e):
    color = RED if e["type"] == "alien" else GREY
    pygame.draw.rect(screen, color, (e["x"] - 18, e["y"] - 18, 36, 36))

def draw_hud(score, time_left):
    screen.blit(font.render(f"Score: {score}", True, (255, 255, 255)), (10, 10))
    screen.blit(font.render(f"Time: {time_left}", True, (255, 255, 255)), (WIDTH -
150, 10))

def get_aimed_enemy(ship_x):
    for e in enemies:
        if abs(ship_x - e["x"]) < 22:
            return e
    return None

ship_x = CENTER_X
ship_y = HEIGHT - 100
score = 0
start_time = pygame.time.get_ticks()
GAME_DURATION = 30000

running = True
while running:
    elapsed = pygame.time.get_ticks() - start_time

```

```

time_left = max(0, (GAME_DURATION - elapsed) // 1000)
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
    if event.type == pygame.KEYDOWN and event.key == pygame.K_SPACE:
        hit = get_aimed_enemy(ship_x)
        if hit:
            score = score + hit["points"]
            new = new_enemy()
            hit["x"], hit["type"], hit["points"] = new["x"], new["type"],
new["points"]

keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT] and ship_x > 100:
    ship_x -= 6
if keys[pygame.K_RIGHT] and ship_x < WIDTH - 100:
    ship_x += 6

screen.fill((10, 12, 40))
for e in enemies:
    draw_enemy(e)
pygame.draw.rect(screen, (80, 220, 255), (ship_x - 20, ship_y, 40, 50))
pygame.draw.polygon(screen, (255, 255, 255), [(ship_x, ship_y - 20), (ship_x -
20, ship_y), (ship_x + 20, ship_y)])
aimed = get_aimed_enemy(ship_x)
laser_color = (255, 210, 40) if aimed else (230, 60, 60)
pygame.draw.line(screen, laser_color, (ship_x, ship_y - 20), (ship_x, 120), 4)
draw_hud(score, time_left)
pygame.display.flip()
clock.tick(60)

pygame.quit()

```

What does `new_enemy()` return, and which keys does the dictionary have?

The `draw_hud` function shows two things. What are they, and where on the screen does each one go?

What does `get_aimed_enemy(ship_x)` do, and what does it return when nothing is lined up?

When you hit an enemy with SPACE, two things happen to it: the score changes and the enemy is replaced. Explain in your own words.

The laser line changes color. When is it yellow, and when is it red?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own version of this game. Make a short phone video where you **explain the code you wrote**. You can show the game running while you talk. Try to use these words: **HUD, score, timer, dictionary, enemy type**.

1. Show the HUD with score and time changing.
2. Read out loud the line that renders the score, and say what it does.
3. Hit an alien and a meteor, and say which gives more points and why.
4. Show what happens when you miss, and explain why it does not crash.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.